SCRATCH – PRELIMINARY, ONE SEED EACH – NOT PART OF THE FINAL REPORT

Report IX: does the completion signal need the rich path_union mix, and do endpoints matter?

Generated July 26, 2026

This is a throwaway working document, not the Report IX manuscript. Every experiment here is a single seed (seed 1000) – exactly why these results live here and not in the real report: we always want several seeds before trusting a number as a real result, and these are not yet replicated. Nothing here should be copied into the real Report IX manuscript until more seeds are in.

What this is

Three new training runs, one seed each, $n = 46$, similarity read-out, RoBERTa-faithful $L = 2$ single-head $d_{\mathrm{model}} = 512$, 10^6 online steps, batch 1000 – identical setup to every other training in this report, only the training *distribution* changes:

- **2-chains-only**: every training sample is a random two-way split of the 46 nodes into two disjoint paths (the short-component size drawn uniformly 1..45 every sample) – never the richer 1–4-path path_union mixture used everywhere else in this project since Report VI.
- **1+2-path**: every sample is, with equal probability, either a single 46-node chain (the k=1 case) or the same random two-way split as above – a narrower mixture than path_union (which also draws 3- and 4-component splits), spanning exactly the two cases path_union itself draws most often.
- **2-cycles-only**: the same random two-way split as the first condition, but each segment closed into a cycle instead of left as an open path (so neither component has a degree-1 endpoint) – the direct training-side analogue of Report VIII's two-cycles test, which only ever evaluated a model that had *never* seen a cycle in training.

All three are evaluated in-distribution with the same split-sweep battery used throughout this project: the behavioural sweep (exact match, reach, cut, predicted-positive rate), the raw pre-bias pre-scale similarity logit by pair type, the real attention scores/α, and the five-stage layerwise similarity geometry ($Z = \mathrm{scale} \cdot \cos + \mathrm{bias}$, connected > 0/disconnected < 0 at every stage). For the two chain-based conditions the sweep runs $a = 1 \,..\, 23$; for the cycle condition it starts at $a = 3$ (a cycle needs at least 3 nodes).

1 2-chains-only training

Result: the endpoint-completion signal reproduces exactly, with a cleaner cut than path_union. Exact match is 1.000 for every split $a \leq 11$, then collapses to 0.000 at *exactly* $a = 12$ – the same break point already seen for the full path_union mixture at this canvas size (§sec:planpre). Reach in the long component drops to 0.79 at the break and then recovers smoothly to 0.96 by the balanced split, closely mirroring path_union's own recovery shape. The one clear difference: **cut stays at** 1.000 **across the entire sweep, never dropping below** 0.96 **even at the fully balanced split** – path_union-trained models degrade to $\approx$ 0.63 cut at the same canvas (§sec:planA). A single, narrow, explicitly-named training distribution reproduces the same completion signal the richer 1–4-path mixture does, and if anything with a more robust cut.

a	exact	reach(long)	reach(short)	cut
1–11	1.000	1.000	1.000	1.000
12	**0.000**	**0.786**	1.000	1.000
15	0.000	0.832	1.000	1.000
18	0.000	0.881	1.000	1.000
20	0.000	0.914	0.995	1.000
23	0.000	0.961	0.961	1.000

Table 1: Split sweep $(a, 46-a)$, seed 1000 only, trained ONLY on random two-chain splits. Selected rows ($a = 1..11$ collapsed since all identical); full sweep in the figure below.

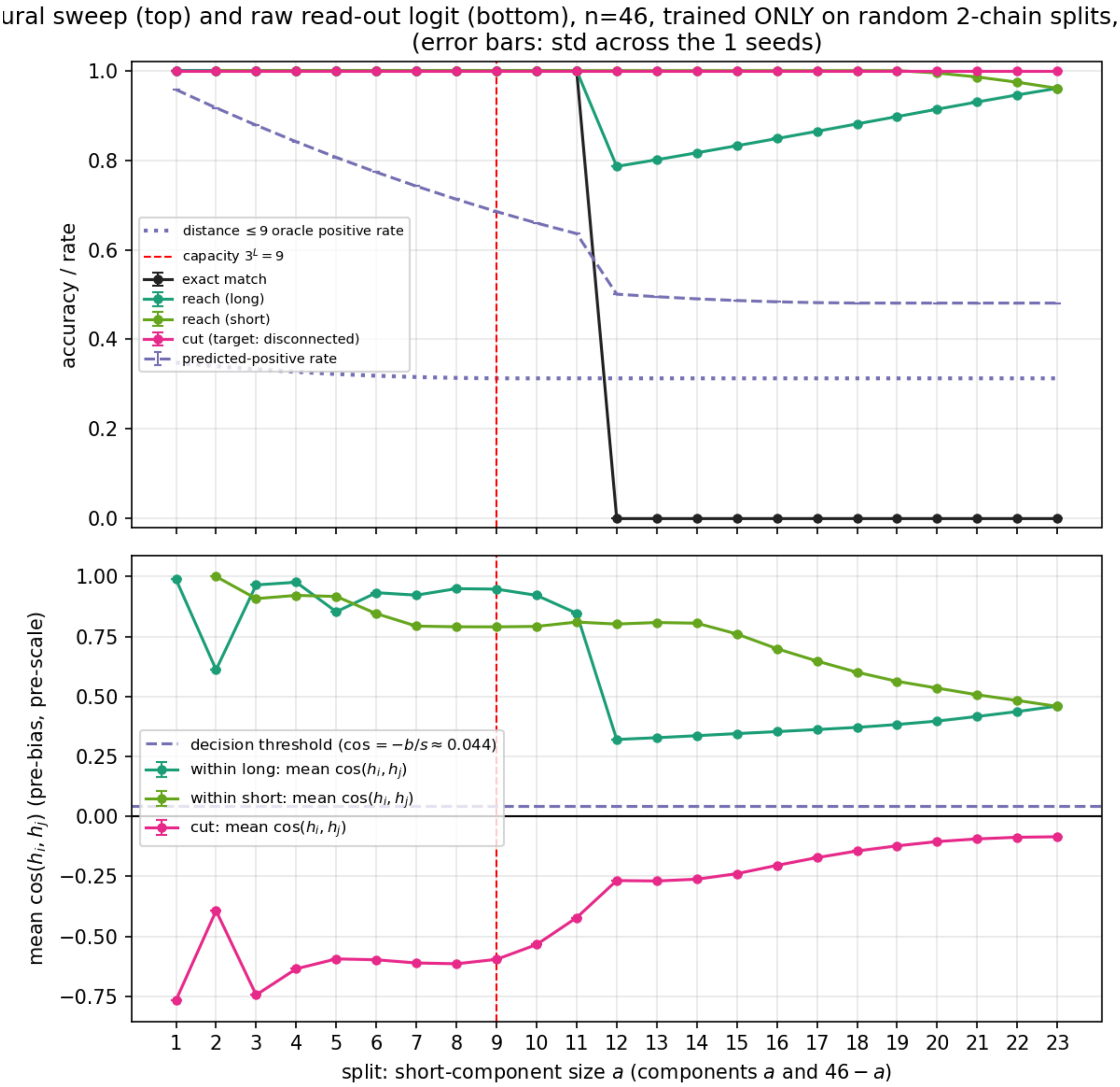

Figure 1: Full sweep + raw similarity logit, seed 1000, trained only on random two-chain splits.

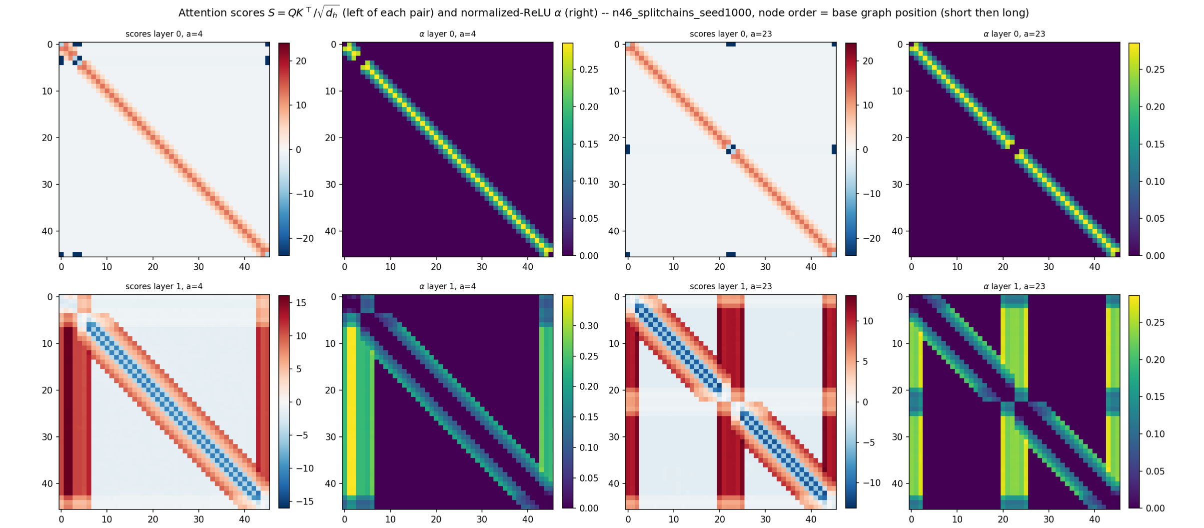

Figure 2: Real attention scores S and normalised-ReLU α, layer 0 (top) / layer 1 (bottom), $a = 4$ and $a = 23$, seed 1000.

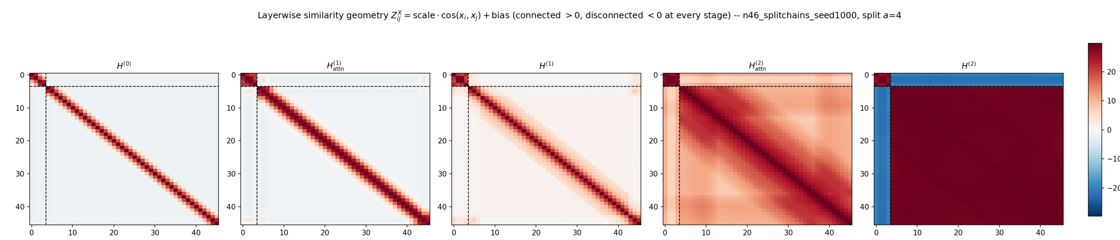

Figure 3: Five-stage similarity geometry, split $a = 4$ (solved, well below the break), seed 1000, trained only on random two-chain splits.

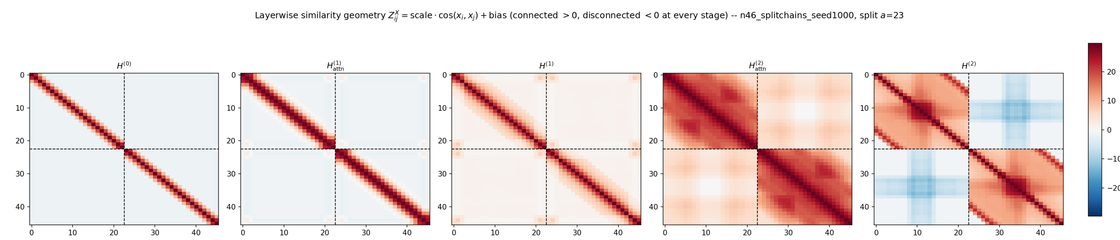

Figure 4: Five-stage similarity geometry, split $a = 23$ (balanced), seed 1000, trained only on random two-chain splits.

2 1+2-path training

Result: essentially identical to 2-chains-only. Adding the single-chain case (k=1) to the mixture changes almost nothing: exact match again breaks at exactly $a = 12$, reach(long) again drops to 0.77 and recovers to 0.92 by the balanced split, and cut again stays high throughout (≥ 0.95 everywhere, only mildly softer than the 2-chains-only condition's flat 1.000 at the most balanced splits). The completion signal is robust to which of the two narrower mixtures is used.

a	exact	reach(long)	reach(short)	cut
1–11	1.000	1.000	1.000	1.000
12	**0.000**	**0.767**	1.000	1.000
15	0.000	0.801	1.000	0.998
18	0.000	0.845	0.989	0.997
20	0.000	0.876	0.967	0.975
23	0.000	0.924	0.922	0.954

Table 2: Split sweep $(a, 46-a)$, seed 1000 only, trained on a 50/50 mixture of a single 46-node chain and random two-chain splits.

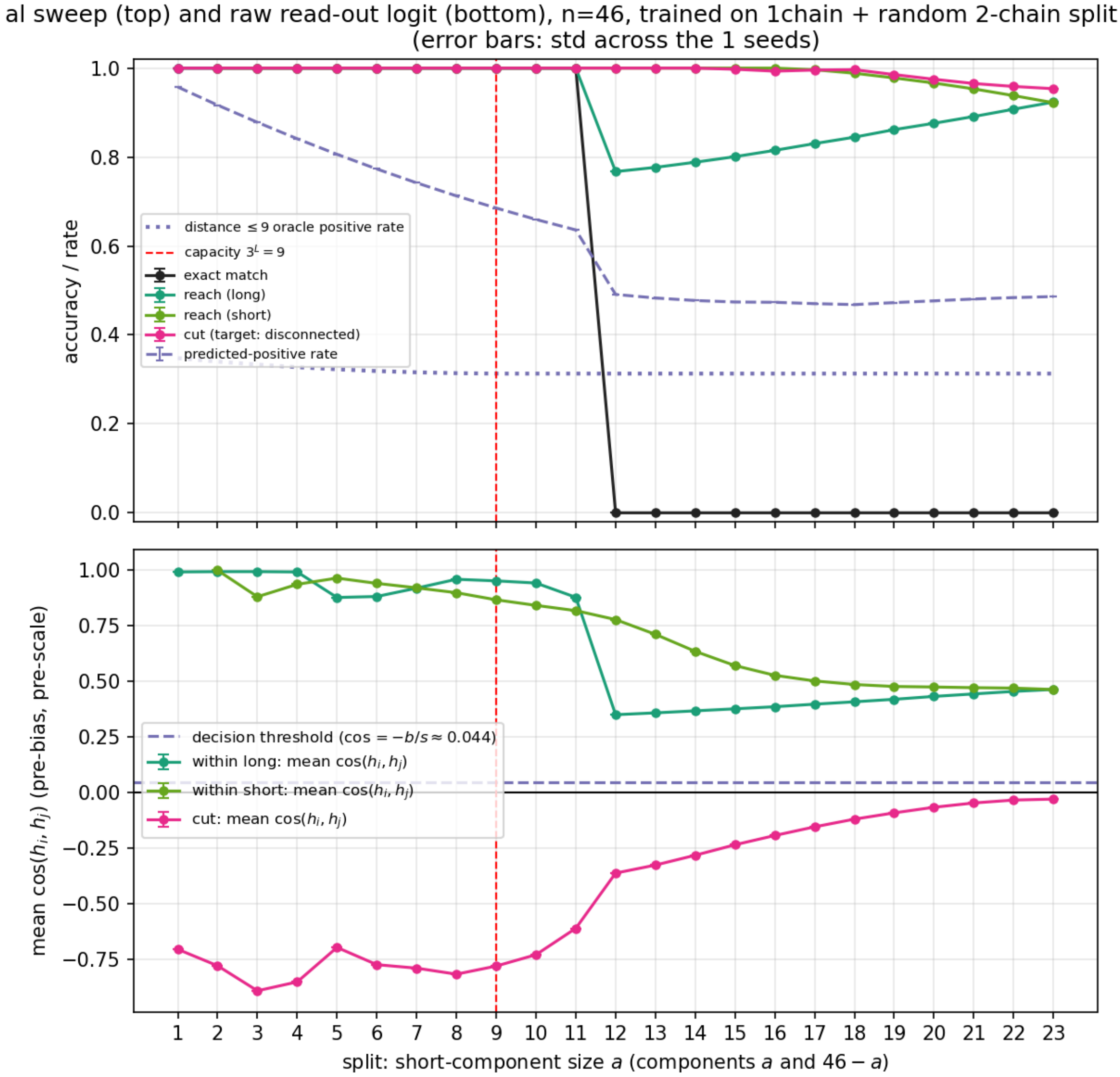

Figure 5: Full sweep + raw similarity logit, seed 1000, trained on 1chain+split_chains.

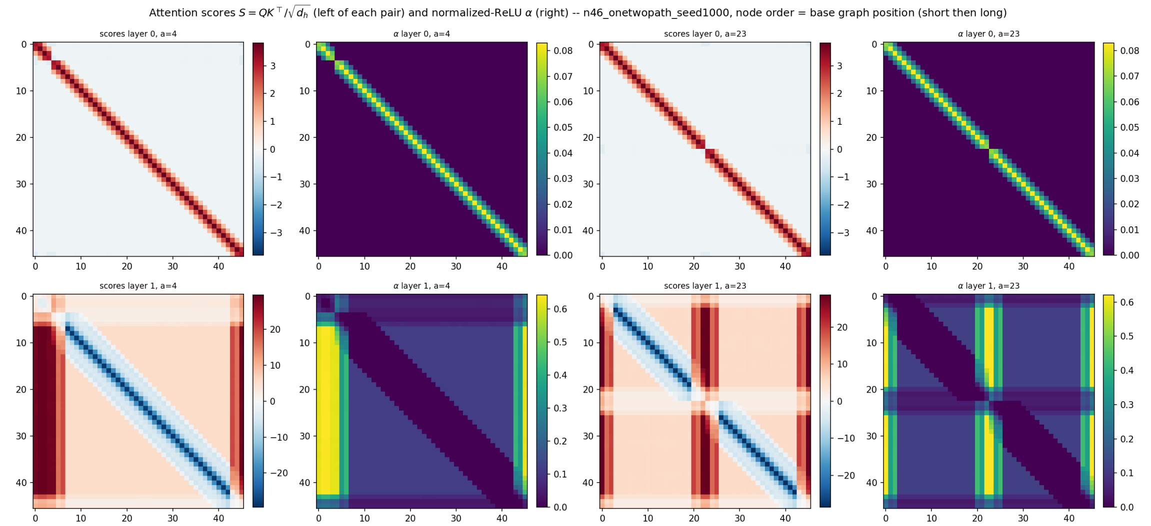

Figure 6: Real attention scores S and normalised-ReLU α, layer 0 (top) / layer 1 (bottom), $a = 4$ and $a = 23$, seed 1000.

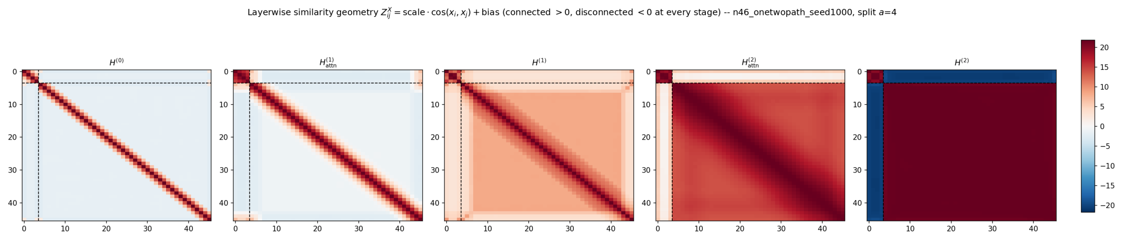

Figure 7: Five-stage similarity geometry, split $a = 4$ (solved, well below the break), seed 1000, trained on 1chain+split_chains.

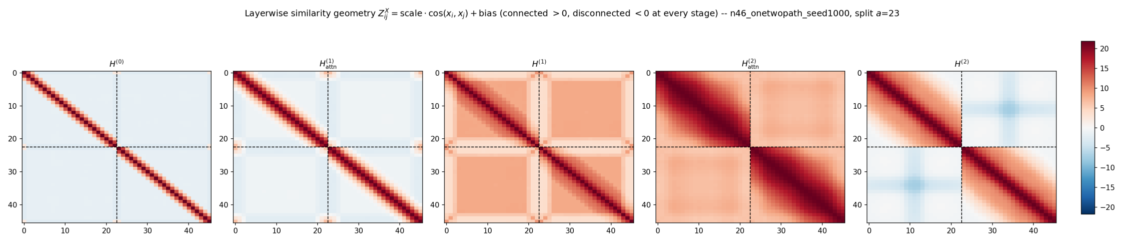

Figure 8: Five-stage similarity geometry, split $a = 23$ (balanced), seed 1000, trained on 1chain+split_chains.

3 2-cycles-only training — a genuinely different failure mode

Result: reach stays perfect everywhere; cut collapses totally and sharply, unlike every chain-based condition. Trained *specifically and systematically* on two-cycle splits (removing Report VIII's "never seen a cycle at all" confound entirely), the model shows a strikingly different signature from every chain-based condition above:

- Reach inside both components is *exactly* 1.000 at *every* split tested, $a = 6$ through $a = 23$ – no dip at the break point at all, unlike every chain condition (which all show a ~ 0.2 dip in reach(long) right at the break).
- Cut is ≥ 0.94 for $a \leq 11$, then collapses to *exactly* 0.000 at $a = 12$ and stays *exactly* 0.000 all the way to the balanced split $a = 23$ – not a gradual softening like the chain conditions, a complete, permanent switch to calling every cross-cycle pair connected.
- One honest caveat: the smallest splits ($a = 3, 4, 5$) are noisy (exact $0.78, 0.51, 0.375$) before the sweep stabilises to 1.000 at $a = 6$–11 – a single seed, so this could be sampling noise around an unusually small/degenerate cycle size, not necessarily a real effect; not investigated further here.

a	exact	reach(long)	reach(short)	cut	pred. positive rate
3	0.780	1.000	1.000	0.985	0.880
4	0.510	1.000	1.000	0.938	0.851
5	0.375	1.000	1.000	0.911	0.824
6–11 (avg)	1.000	1.000	1.000	1.000	0.701
12	**0.000**	**1.000**	**1.000**	**0.000**	**1.000**
15	0.000	1.000	1.000	0.000	1.000
18	0.000	1.000	1.000	0.000	1.000
23	0.000	1.000	1.000	0.000	1.000

Table 3: Split sweep $(a, 46-a)$, seed 1000 only, trained ONLY on random two-cycle splits (each segment closed into a ring). Note reach is always exact; cut drops to exactly 0, not a soft floor. **The predicted-positive rate is EXACTLY** 1.000 **from** $a = 12$ **onward**: past the break, the model predicts every single pair in the graph (all n^2 of them, not just the cross pairs) as "connected" – not a selective cut failure, but total collapse to "yes". Since within-component pairs are trivially correctly "connected" regardless, a 2-component graph alone cannot tell this apart from the model having learned to selectively recognise the small component and correctly separate it – a 3-cycle test (pending, see below once the job returns) is needed for a graph that CAN tell the two apart.

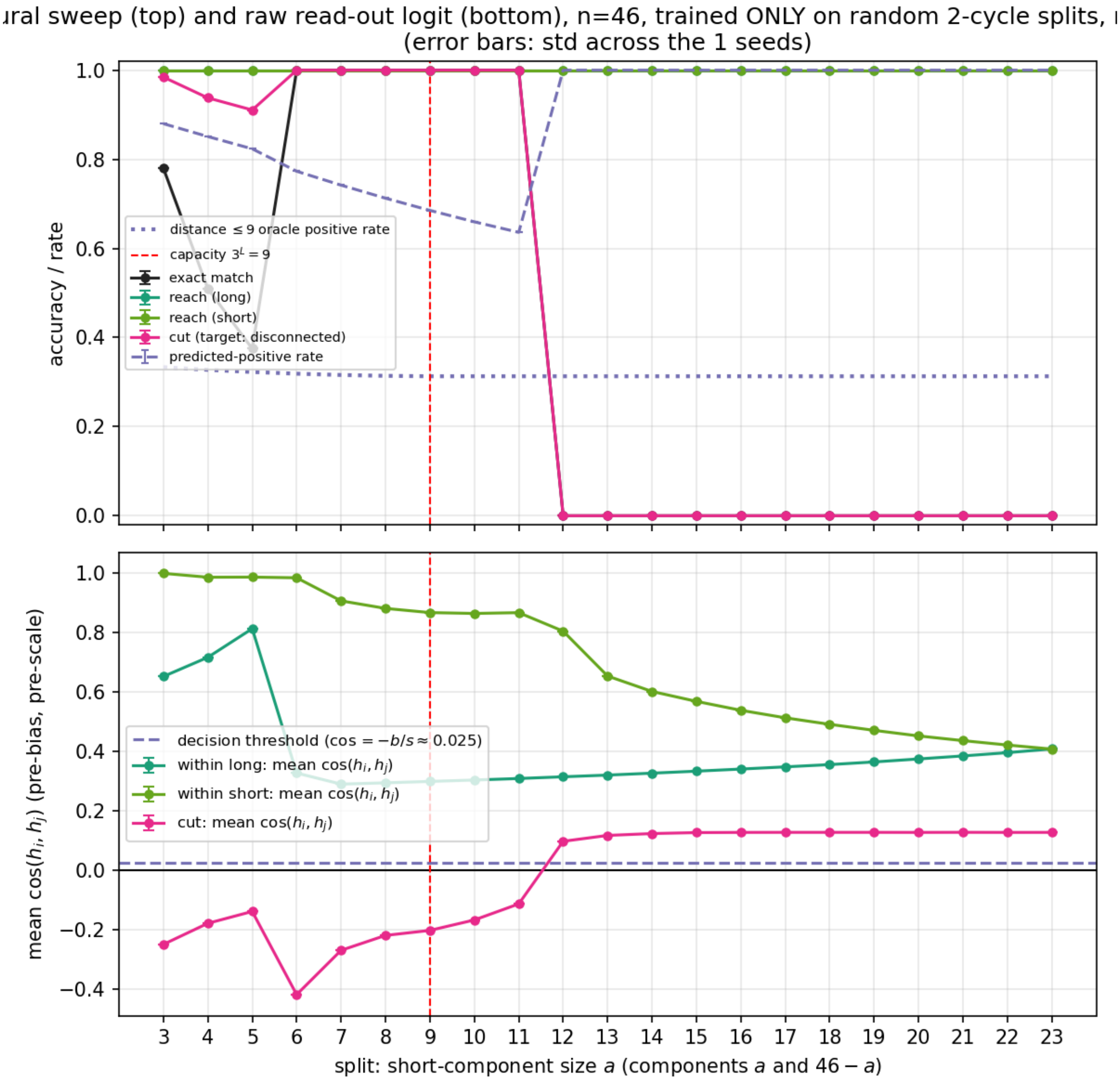

Figure 9: Full sweep + raw similarity logit, seed 1000, trained only on random two-cycle splits. Note the cut curve (pink) hits exactly 0 at $a = 12$ and never recovers, unlike the gradual softening in the two chain-based conditions above.

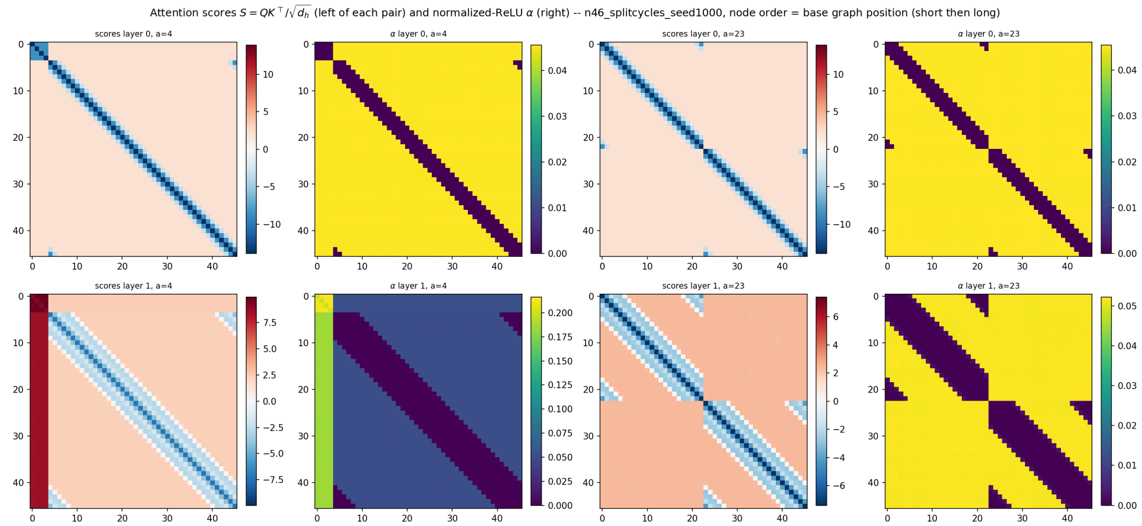

Figure 10: Real attention scores S and normalised-ReLU α, layer 0 (top) / layer 1 (bottom), $a = 4$ and $a = 23$, seed 1000. Layer 1 α is far more uniformly high across almost the entire matrix at $a = 23$ than in either chain-based condition, where the high-α region was concentrated in bands connecting specific components rather than nearly everywhere.

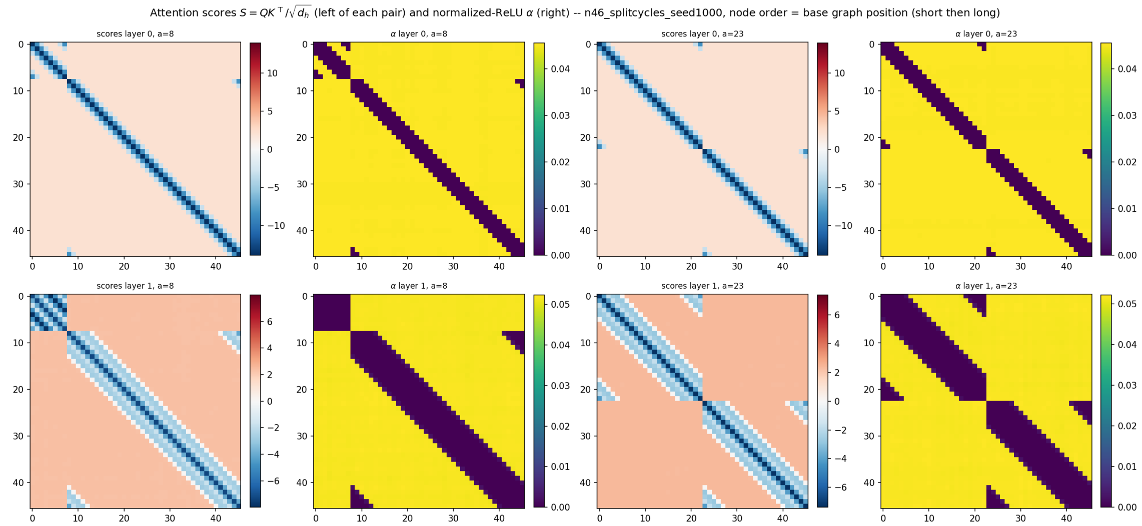

Figure 11: Same, but $a = 8$ vs $a = 23$ – $a = 8$ is a more honest "solved" reference point than $a = 4$ above, since it sits inside the clean $a = 6$–11 plateau (exact match = 1.000) rather than the borderline $a = 4$ cell (exact match only 0.51).

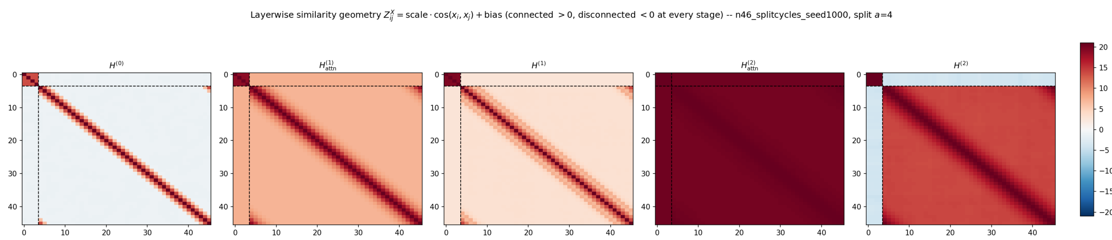

Figure 12: Five-stage similarity geometry, split $a = 4$ (borderline, exact = 0.51), seed 1000.

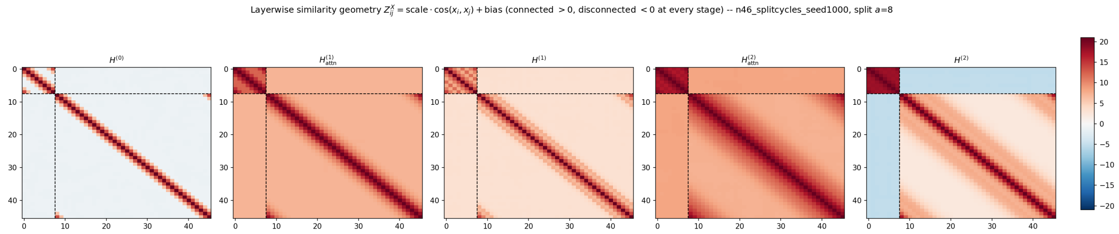

Figure 13: Five-stage similarity geometry, split $a = 8$ (cleanly solved, exact = 1.000, inside the $a = 6$–11 plateau), seed 1000 – the more honest "before the break" reference point.

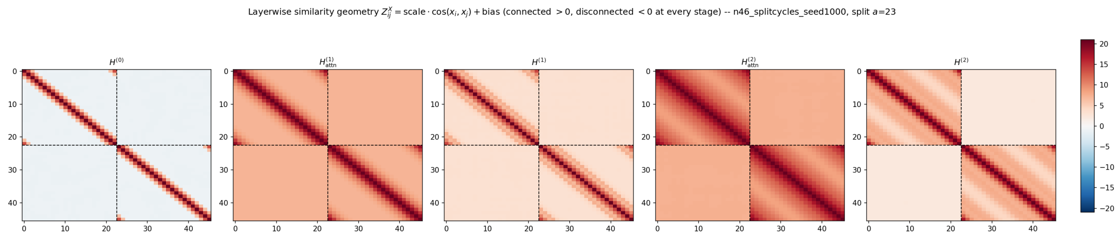

Figure 14: Five-stage similarity geometry, split $a = 23$ (balanced, totally failed), seed 1000. Note the cross-component block is already mildly *positive* (pale red, not blue) at $H^{(0)}$ – before any attention runs at all – and never turns negative at any later stage, unlike the chain-based conditions where the cross block starts clearly negative at $H^{(0)}$.

A candidate mechanistic reading (tentative, one seed, one representative split). In the balanced-split heatmap above, the cross-component block is already faintly on the "connected" side of zero at the very first stage $H^{(0)}$ (the read-in embedding, before any attention or feed-forward computation) and stays that way through every later stage — it never crosses into clearly negative territory the way it does for the chain-based conditions. Since $H^{(0)}$ is built purely from each node's own row of the adjacency matrix, and every node on a cycle has an *identical* local neighbourhood shape (degree exactly 2, no distinguished end), this is consistent with the read-in step itself failing to give cycle nodes from different components a distinguishing signature the way a path's endpoint-distance does for chain nodes — before the network has any chance to build a genuine cut decision downstream. This is offered as a candidate explanation to test further (e.g. by looking at $H^{(0)}$ across more splits and seeds), not a settled conclusion.

4 Thread A.3: does the split_chains-only checkpoint survive a genuine third component? All 176 size combinations

The 2-chains-only checkpoint from §1 was trained EXCLUSIVELY on graphs with exactly 2 disjoint path components, never 3 or more – so a genuine three-component graph is out-of-distribution in component *count* for this checkpoint specifically (unlike the three-component tests of Reports VII/VIII, whose training streams already covered 3–4 components). This is the direct implementation of Thread A.3 of the plan: take the same two-way split and remove one internal edge from one of the two paths, producing 3 disjoint paths with sizes the training stream could never have produced. Every distinct combination of resulting sizes was tested – every partition of $n = 46$ into 3 positive parts ($s_1 \leq s_2 \leq s_3$), 176 cells in total, 200 random relabellings per cell – purely behavioural for now (exact match, reach inside each component, cut between each of the three pairs); no attention scores or layerwise geometry yet, pending a choice of which specific cells are worth the deeper dive.

Headline 1: whole-graph exact match is 0.000 **in every single one of the** 176 **cells.** Introducing a genuine third component breaks perfect exact-match completely, with no exceptions – even the easiest-looking configurations (e.g. a trivial isolated node plus a short path plus a long one) never get every pair in the graph right simultaneously.

Headline 2: reach stays essentially perfect everywhere; the whole story is which ONE of the three pairwise cuts fails. Reach inside each individual component is ≥ 0.95 in nearly every cell regardless of size (not tabulated in full here since it is uniformly high) – the model still resolves distance correctly *inside* each component even with a foreign third component present. What fails is entirely at the cut level, and remarkably it is usually not a generic "confused everywhere" failure: in most cells exactly ONE of the three pairwise relationships (comp_1-comp_2, comp_1-comp_3, comp_2-comp_3) is called "connected" almost always, while the other two are correctly called "disconnected" almost always.

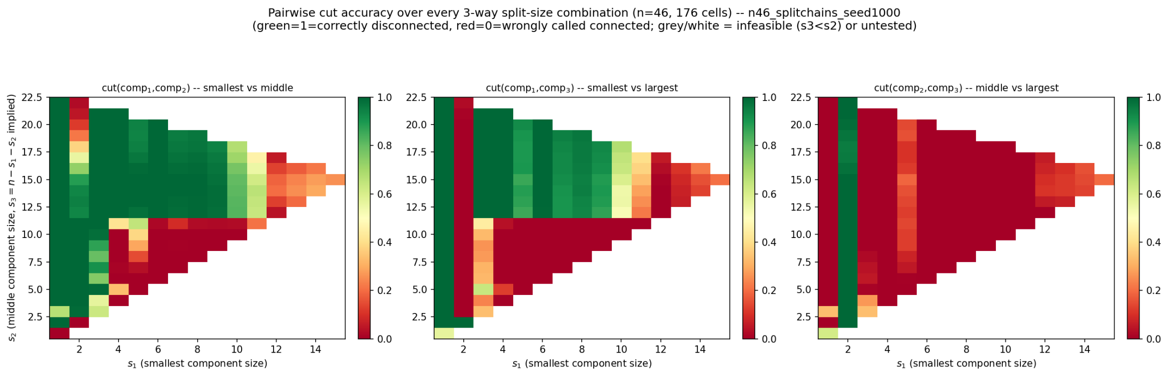

Figure 15: Each of the three pairwise cut accuracies, over every feasible (s_1, s_2) combination ($s_3 = 46 - s_1 - s_2$ implied), green=correctly disconnected, red=wrongly called connected.

Headline 3: a component of exactly 1 **node (an isolated node, the training distribution's own "short" extreme) is the ONLY size that lets the model tell the other two components apart from each other.** The rightmost panel above (cut between the middle and largest component) is red – failed – across almost the *entire* grid, with one striking exception: a clean green vertical stripe at $s_1 = 1$. The moment the smallest component has any internal structure at all ($s_1 \geq 2$), the model can no longer distinguish the other two real path components from one another, no matter how different their sizes are (e.g. sizes $(4, 12, 30)$

still fail comp$_2$-comp$_3$ cut at 0.000). This directly generalises the Thread A.4 short-vs-short collapse (§sec:planA of the official report) down to $K = 3$: the model's cut mechanism appears to need a genuinely trivial, structureless reference component to anchor against; without one, two "real" components collapse into each other.

Headline 4 (an anomaly worth flagging): $s_1 = 2$ (a single edge, the other degenerate extreme) inverts which pair fails. At $s_1 = 1$ the failing pair is (comp$_2$,comp$_3$); at $s_1 = 2$ it flips – cut(comp$_2$,comp$_3$) mostly SUCCEEDS instead, while cut(comp$_1$,comp$_3$) (smallest vs largest) fails almost everywhere (the middle panel's red stripe at $s_1 = 2$). For every $s_1 \geq 3$ the dominant failing pair reverts to (comp$_2$,comp$_3$), matching the $s_1 = 1$ story. This one-cell flip at $s_1 = 2$ is the single most surprising thing in this sweep and is a natural first candidate for the mechanistic dive.

Headline 5: once nothing is small (all three sizes close together), everything degrades. For $s_1 \gtrsim 12$ (sizes approaching an even three-way split, e.g. $14, 15, 17$ or $15, 15, 16$), all three cuts drop into the 0–0.3 range together – there is no longer even a partial anchor, and the model is uniformly unable to place any of the three cut boundaries correctly.

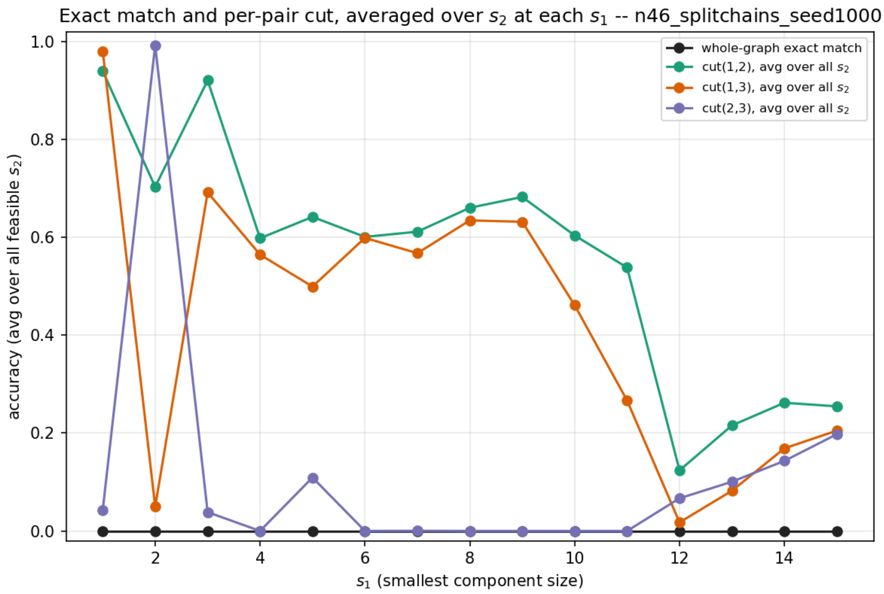

Figure 16: Exact match (always 0) and each pairwise cut, averaged over every feasible s_2 at each s_1. The cut(2,3) trace (purple) is near 1 only at $s_1 = 1$, collapses immediately for every $s_1 \geq 3$, and cut(1,3) (orange) is the one that is near 0 instead at $s_1 = 2$ – the inversion of Headline 4.

Candidate cells for the next step (attention scores / layerwise cosine geometry) – for the user to choose from. A few natural candidates, spanning the qualitatively different regimes above:

- $(s_1, s_2, s_3) = (1, 22, 23)$ – the "everything works except reach/cut noise" case, the cleanest baseline with a trivial isolated node.
- $(s_1, s_2, s_3) = (4, 15, 27)$ or $(7, 15, 24)$ – a typical "2-out-of-3 cuts succeed, cut(2,3) fails" cell, to see the mechanism behind Headline 3 directly.
- $(s_1, s_2, s_3) = (2, 10, 34)$ – the $s_1 = 2$ anomaly (Headline 4), to see whether the inversion shows up as a visibly different attention/geometry pattern from the $s_1 \geq 3$ cells.

- $(s_1, s_2, s_3) = (15, 15, 16)$ – the fully-balanced, everything-fails regime (Headline 5).

4.1 Same test, on the 1chain+split_chains checkpoint: cleaner, but the same core story

The identical 176-cell sweep, unchanged, run against the OTHER n=46 seed-1000 checkpoint that also never saw 3+ components: the §2 checkpoint trained on a 50/50 mixture of a single full chain (k=1) and random two-way splits (k=2). Whole-graph exact match is again 0.000 in all 176 cells.

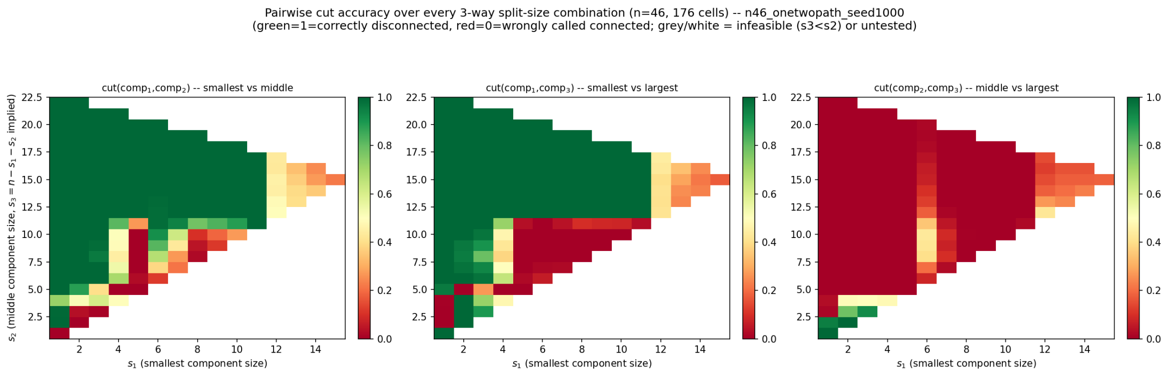

Figure 17: Same three pairwise cut heatmaps as the split_chains-only version above, on the 1chain+split_chains checkpoint.

The core story is the same: cut(2,3) still fails almost everywhere. The dominant pattern is unchanged – cut(comp_2,comp_3) (the two non-smallest components) is red (failed) across almost the entire grid, exactly as for the split_chains-only checkpoint.

But cut(1,2) and cut(1,3) are markedly more robust here. For split_chains-only, cut(1,2)/cut(1,3) averaged (over s_2) only 0.5–0.7 through most of $s_1 = 4$–11 and were visibly noisy cell-to-cell. For 1chain+split_chains, the SAME two cuts average 0.75–1.0 across that entire range and stay high all the way out to $s_1 = 11$ (e.g. cut(1,2)= 0.999 at $s_1 = 11$, vs. only 0.538 for split_chains-only at the same s_1) – a visibly cleaner, wider green region in the heatmap above compared to Figure 15. Seeing single, un-split chains during training (the $k = 1$ case) appears to make the model noticeably better at telling a non-trivial component apart from a small one, even though it still never saw 3 components.

The $s_1 = 2$ anomaly (Headline 4) does NOT replicate here. For split_chains-only, $s_1 = 2$ inverted which pair fails (cut(1,3)$\approx$ 0.05, an outlier). For 1chain+split_chains, $s_1 = 2$ is unremarkable: cut(1,3)= 0.947, in line with its neighbours, no inversion. This suggests the $s_1 = 2$ anomaly is a property of that specific checkpoint/seed, not a general structural rule about single-edge components.

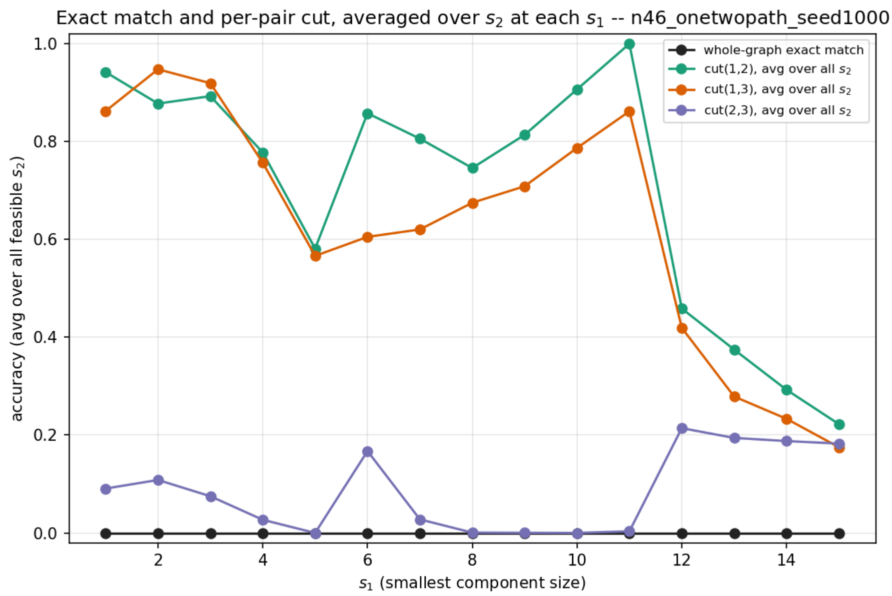

Figure 18: Same by-s_1 summary as the split_chains-only version above, on the 1chain+split_chains checkpoint. Note cut(1,2)/cut(1,3) (green/orange) sit visibly higher and smoother across $s_1 = 1$–11 than the split_chains-only version, and there is no dip at $s_1 = 2$.

Balanced-split degradation (Headline 5) still holds. For $s_1 \gtrsim 12$ all three cuts still collapse together here too (e.g. cut(1,2) drops from 0.999 at $s_1 = 11$ to 0.459 at $s_1 = 12$), just as for split_chains-only.

Cross-experiment takeaways (one seed each — read directionally, not definitively)

1. **The endpoint-completion signal does not need the rich path_union mixture.** Both narrower chain-based distributions (2-chains-only, 1+2-path) reproduce the identical break point ($a = 12$) and the same qualitative reach-dip-then-recover shape as the full 1–4-path mixture, with an equal-or-cleaner cut. If anything, the narrower distributions are *more* robust on cut, not less — the opposite of what one might expect if the phenomenon depended on having seen a wide variety of component counts.
2. **Cycles, trained on properly, are NOT simply another counterexample to "the diameter is the difficulty measure" in the same way chains are.** They show a *cleaner reach* (perfect at every split, no dip at all) but a *much harsher cut failure* (exact, permanent 0 rather than a soft floor) once the split is large enough. This is a genuinely different failure signature from every chain-based condition, not a milder or stronger version of the same one.
3. **This speaks directly to the endpoints question.** Report VIII's two-cycles test could not distinguish "cycles are unlearnable" from "this model never saw a cycle at all." Training specifically and systematically on cycles removes that confound, and the cut *still* fails totally past a threshold split – so the earlier OOD collapse was not purely an unfamiliarity artefact. Whether this means path endpoints are strictly *necessary* for the cut-sharpening mechanism, or whether cycles are simply harder in some other structural sense (every node locally identical, no distinguishing feature at the read-in step at all – the candidate reading above), is not yet settled by a single seed and needs more seeds and probably more splits between $a = 5$ and $a = 12$ to pin down exactly where cut starts to go wrong.

What is still missing before any of this becomes real Report IX content

- More seeds for all three conditions (this document is one seed each, by design — see the warning box at the top).
- For the 2-cycles condition specifically: a denser split sweep between $a = 5$ and $a = 12$ to locate the cut break point precisely (currently only $a = 3, 4, 5, 6, \ldots, 11, 12$ are covered, with a suspicious dip at $a = 3$–5 worth a second look), and the candidate read-in-geometry explanation should be checked systematically (more splits, more seeds) before it is stated as anything more than a hypothesis.
- Only after that: fold the confirmed, multi-seed picture into the real Report IX manuscript (Thread A.2/A.5-adjacent for the chain-based results, Thread B for the cycles result).
- For Thread A.3 (the three-way split sweep): still purely behavioural on one seed – attention scores and layerwise cosine geometry are the deliberate next step, on whichever candidate cells above turn out to be of interest, before this becomes Report IX content in its own right.